

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

4.3 – Solution Architecture

"Plugging into the Future: An Exploration of Electricity Consumption Patterns" outlines an end-to-end Solution Architecture designed to monitor, analyse, and predict energy usage through an integrated hardware-and-software ecosystem. Grounded in recent implementations, this framework leverages Internet of Things (IoT) hardware, robust data-pipelining, predictive machine learning, and dynamic user interfaces to dramatically lower energy waste and optimize load management. [1, 2, 3]

The technical breakdown below details the four essential tiers of this modern smart-grid and facility-level solution architecture. [2, 4, 5]

TIER 1: The Edge & Data Collection Layer (IoT Layer)

This layer acts as the physical entry point, capturing real-time metrics across residential, commercial, or institutional grids. [2, 3, 6, 7]

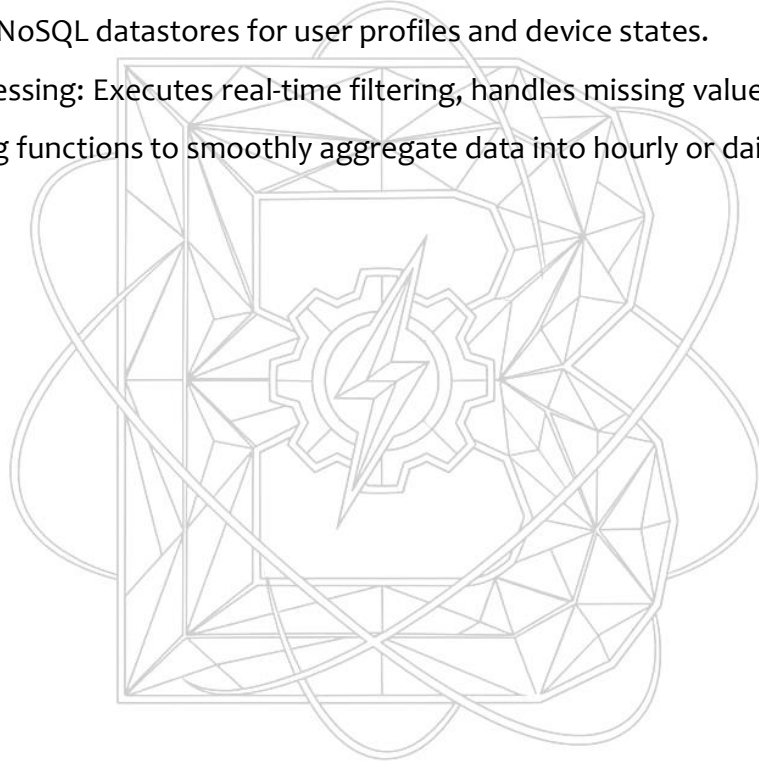
- Smart Infrastructure: Deploys localized hardware like [smart plugs](#) and advanced metering systems directly onto high-load appliances and circuits. [1, 8]
- Telemetry Collection: Continuously extracts real-time vital signs including voltage, current, active power (kW), total consumption (kWh), and transient spikes. [8, 9]
- Edge Communication: Transmits data payloads through lightweight protocols like MQTT or CoAP over Wi-Fi, Zigbee, or cellular networks to a central cloud gateway. [9, 10, 11]

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TIER 2: The Core Data Pipeline & Storage Layer

This operational tier securely ingests, formats, and stores volatile streaming data. [2, 12]

- Ingestion Engines: Leverages messaging brokers (such as Apache Kafka or AWS IoT Core) to handle highly concurrent incoming telemetry without data drops. [13]
- Storage Strategy: Employs a hybrid approach utilizing Time-Series Databases (e.g., Influx DB or Timescale DB) for rapid indexing of sequential sensor logs alongside relational/NoSQL datastores for user profiles and device states.
- Data Processing: Executes real-time filtering, handles missing values, and implements windowing functions to smoothly aggregate data into hourly or daily intervals. [14, 15, 16, 17].



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